

# PilotFish OCI Object Storage Transport – Complete Reference (Custom Sandbox Module)

Derived from Sandbox custom module source (not in 26R1.11 WAR)

August 4, 2026

## At a glance (plain English)

**Custom module (Sandbox)** — not shipped in the standard eiPlatform WAR. This transport **uploads the transaction body as one object** to Oracle Cloud Infrastructure Object Storage using a signed **PutObject** call (OCI Java SDK). Useful for cloud archival or feeding bucket-triggered downstream systems.

## OCI Object Storage Transport (Custom)

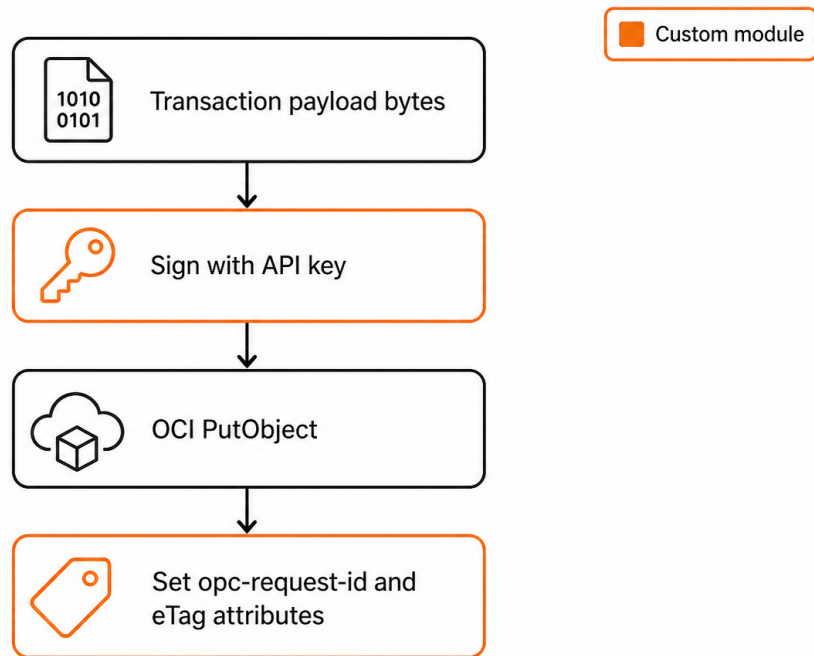


Figure 1: At a glance (plain English)

- **Uploads** payload bytes to a configured bucket and object name (OGNL supported).
- **Authenticates** with API Key (tenancy, user, fingerprint, PEM private key).
- **Supports** optional **Service Endpoint Override** for local emulators (e.g. floci-oci).

- **Records** opc-request-id, ETag, and object metadata on the transaction.

**Provenance:** Sandbox demo module at `PilotFish Sandbox/Clients/Demos/edi-835-oci-bucket/custom`.  
Deploy as a custom JAR — not present in `eip.war.hs.26R1.11`.

## Purpose

This document is a **deep dive** into the **OCI Object Storage** custom transport: PutObject execution, credential resolution, registered attributes, and configuration — authored from source because the module is **not** in the 26R1.11 WAR decompile tree.

| Provenance            | Value                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------|
| WAR                   | <b>N/A — custom module</b>                                                                              |
| Source                | .../oci-object-storage-transport/src/main/                                                              |
| Sandbox context       | PilotFish<br>Sandbox/Clients/Demos/edi-835-oci-bucket/                                                  |
| Transport type string | OCI Object Storage                                                                                      |
| Description           | Put objects to Oracle Cloud<br>Infrastructure Object Storage<br>(signed PutObject via OCI<br>Java SDK). |
| Categories            | WEB, SPECIAL_PROTOCOL                                                                                   |
| Dependencies          | Oracle OCI Java SDK<br>(ObjectStorageClient,<br>SimpleAuthenticationDetailsProvider)                    |

## 1. Conceptual model

Unlike filesystem or JDBC transports in the core WAR, this module is a **thin PutObject wrapper**: read entire transaction stream to memory, sign request with API key material, upload, stamp response metadata as attributes.

```
Transaction payload (bytes)
|
v
Resolve config (region, namespace, bucket, object name, content type)
|
v
Build SimpleAuthenticationDetailsProvider (API key PEM)
|
v
ObjectStorageClient [optional setEndpoint override]
|
v
PutObjectRequest -> putObject
|
v
Set com.pilotfish.oci.* attributes
```

No download/list/delete operations in this demo module.

## 2. High-level behavior (executeTransport)

From `OciObjectStorageTransport.executeTransport`:

1. **Required strings:** Region, Namespace, Bucket, Object Name, Tenancy OCID, User OCID, Fingerprint (trimmed; blank → `EIException`).
2. **Content-Type:** default `application/octet-stream` if blank; config default in descriptor is `application/json`.
3. **Private key:** inline PEM (`OciPrivateKeyPem`, `\n` escape supported) **or** read file from `OciPrivateKeyPath`; both empty → error.
4. **Read body:** `IOUtils.toByteArray(data.getDataStream())`.
5. **Auth builder:** tenancy, user, fingerprint, `StringPrivateKeySupplier`, `Region.fromRegionId`.
6. **Client:** build, optionally `setEndpoint` when override non-blank (emulator / dedicated endpoint).
7. **PutObject** with content length, content type, body stream.
8. **Attributes:** `opc-request-id`, `ETag`, object name, bucket, namespace.
9. **Errors:** `BmcException` includes HTTP status in message; client closed in **finally**.

Standard `AbstractTransport.sendData` wraps with complete/error states.

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## 3. Configuration reference

| UI label                     | Tag                            | Type   | Default                       | Tab         | Required   |
|------------------------------|--------------------------------|--------|-------------------------------|-------------|------------|
| Region                       | <code>OciRegion</code>         | String | <code>us-ashburn-1</code>     | Basic       | Yes        |
| Namespace                    | <code>OciNamespace</code>      | String | <code>""</code>               | Basic       | Yes        |
| Bucket                       | <code>OciBucket</code>         | String | <code>""</code>               | Basic       | Yes        |
| Object Name                  | <code>OciObjectName</code>     | String | <code>""</code>               | Basic       | Yes (OGNL) |
| Content Type                 | <code>OciContentType</code>    | String | <code>application/json</code> | Advanced    | No         |
| Service Endpoint<br>Override | <code>OciEndpoint</code>       | String | <code>""</code>               | Advanced    | No         |
| Auth Mode                    | <code>OciAuthMode</code>       | Choice | API Key                       | Credentials | —          |
| Tenancy OCID                 | <code>OciTenancyOcId</code>    | String | <code>""</code>               | Credentials | Yes        |
| User OCID                    | <code>OciUserOciId</code>      | String | <code>""</code>               | Credentials | Yes        |
| Fingerprint                  | <code>OciFingerprint</code>    | String | <code>""</code>               | Credentials | Yes        |
| Private Key File             | <code>OciPrivateKeyPath</code> | String | <code>""</code>               | Credentials | No*        |
| Private Key PEM<br>(inline)  | <code>OciPrivateKeyPem</code>  | String | <code>""</code>               | Credentials | No*        |

\* One of private key file or inline PEM required at runtime.

**Auth Mode** currently offers only **API Key** in the demo module.

### 3.1 Registered transaction attributes

| Attribute key                             | Type          | Description         |
|-------------------------------------------|---------------|---------------------|
| <code>com.pilotfish.oci.requestId</code>  | <b>String</b> | OCI opc-request-id  |
| <code>com.pilotfish.oci.eTag</code>       | <b>String</b> | ETag from PutObject |
| <code>com.pilotfish.oci.objectName</code> | <b>String</b> | Object written      |
| <code>com.pilotfish.oci.bucket</code>     | <b>String</b> | Bucket written      |
| <code>com.pilotfish.oci.namespace</code>  | <b>String</b> | Namespace written   |

Metadata registered with `TransactionAttributesMetadata.LISTENER` scope in source.

### 3.2 Inherited AbstractTransport

Uses standard transport lifecycle; no additional inherited options beyond base module.

## 4. Worked example — emulator upload

Demo configuration (from source comment):

- **Service Endpoint Override:** `http://floci-oci:4599`
- **Region:** `us-ashburn-1`
- **Namespace / Bucket / Object Name:** from sandbox bucket setup
- **Private Key File:** path to API signing key PEM

Transaction JSON payload uploads to emulator; attributes capture request id for correlation in logs.

## 5. Known quirks and caveats

1. **Custom / not in WAR** — must build and install JAR via eiPlatform custom module path.
2. **PutObject only** — no multipart, append, or server-side encryption options in source.
3. **Full memory read** — entire transaction loaded to byte array before upload.
4. **Inline PEM in config** — demo/dev convenience; prefer file path in production.
5. **Single auth mode** — resource principals / instance principals not implemented.
6. **Object name OGNL** — resolved via standard configuration manager string evaluation.

## 6. Operational recommendations

| Goal       | Guidance                                                      |
|------------|---------------------------------------------------------------|
| Deployment | Package with OCI SDK dependencies aligned to sandbox POM      |
| Secrets    | Use <b>Private Key File</b> on secured volume, not inline PEM |

| Goal       | Guidance                                                                                                        |
|------------|-----------------------------------------------------------------------------------------------------------------|
| Testing    | Point <b>Service Endpoint Override</b> at floci-oci or compatible emulator                                      |
| Monitoring | Log <code>com.pilotfish.oci.requestId</code> for OCI support correlation                                        |
| Versioning | Treat as <b>Custom</b> doc; PDF stored under <code>Documents/Transports/26R1.11/</code> for library consistency |

## 7. Source index (Sandbox)

| Topic                        | Path                                                            |
|------------------------------|-----------------------------------------------------------------|
| OCI Object Storage transport | PilotFish<br>Sandbox/Clients/Demos/edi-835-oci-bucket/custom-m  |
| Related demo doc             | PilotFish<br>Sandbox/Clients/Demos/edi-835-oci-bucket/documents |

## 8. Pipeline provenance

PilotFish Sandbox custom module (NOT eip.war.hs.26R1.11)

-> source-level documentation

-> `Documents/Transports/26R1.11/PilotFish-OCI-Object-Storage-Transport-Reference-Custom.(md|j`

*Deep-dive documentation from Sandbox custom source – OCI Object Storage Transport – August 2026.*